

ARCHITECTING INTELLIGENT INTAKE

Autonomous Agent Triage

Ashford & Kane LLP | Intake Engineering Team

The Setup: Chaos Before Agents

Client intake triage was totally manual, causing massive bottlenecks across Corporate, Family, and Criminal law departments.

- **Manual Triage:** Intake was assigned based on "gut feel," resulting in misrouted cases and wasted billable attorney hours.
- **The Bottleneck:** Urgent legal matters sat in queues while clients waited days for callbacks.
- **The Ethics Risk:** Conflicts of interest weren't checked until *after* an attorney reviewed the file.
- **Our Goal:** Build an automated system to classify practice areas, check conflicts, assess complexity, and route immediately.



Attempt 1: The Reactive Agent

We started with a pure rule-based loop using simple, hard-coded keyword matching and zero API calls or tokens.



The Appeal

It was instant (<5ms latency) and completely free (\$0.00). In controlled environments, it predictably routed cases based on simple keywords.



The Mechanics

Strict "if/then" rules dictated actions. "Divorce" routed to Family Law; "DUI" routed to Criminal Defense.



The Failure

It failed Test 4 entirely by routing a multi-issue case to the wrong area. It only handled conflicts 70% of the time and broke entirely on nuance.

Attempt 2: Unconstrained ReAct

We shifted to a generative model (**Gemini 1.5 Flash**) to reason, select tools, and decide when to stop independently.

- **The Breakthrough:** It handled ambiguous cases nicely, catching 95% of conflicts and 93% of pro-bono routing.
- **The Hallucinations:** The model hallucinated tool names 15% of the time, inventing actions that didn't exist in our code.
- **Infinite Loops:** On Test 4, it completely spun out—calling a tool 5 times and providing an unclear answer.
- **The Cost:** Averaged ~800 tokens and 3.2 API calls per request, causing massive 1200–2500ms latency spikes and costing ~\$0.005.



Attempt 3: Deterministic Routing



To stop the loops, we forced Gemini to make *one* classification call (~150 tokens) into 5 fixed buckets, followed by standard code.

- **Control Regained:** It was incredibly fast (400–800ms) and cheap (~\$0.001 per request), bringing hallucinations down to 5%.
- **Loss of Adaptability:** It only hit 88% accuracy on conflicts and 85% on pro bono cases.
- **The Verdict:** It lacked multi-step reasoning. In Test 4, it completely failed to handle the ambiguity of the dual-issue lawsuit and divorce.

Attempt 4: Constrained ReAct

We brought back the ReAct loop but wrapped Gemini 1.5 Flash in strict guardrails to find the perfect balance.



Schema Validation

Every reasoning step had to pass strict JSON validation. This dropped our hallucination rate exactly to 0%.



Tool Allow-List

We restricted the model to an explicit allow-list of 4 tools, eliminating the risk of rogue code execution.



Hard Budget

We enforced a `MAX_STEPS = 6` limit. On Test 4, the model couldn't loop; it analyzed the data and correctly escalated the issue.

Performance Metrics Comparison

Metric	Reactive	Unconstrained ReAct	Deterministic Routing	Constrained ReAct
Avg API Calls / Tokens	0 / 0	3.2 / ~800	1 / ~150	2.1 / ~400
Latency	<5ms	1200–2500ms	400–800ms	800–1400ms
Cost per Request	Free (\$0.00)	~\$0.005	~\$0.001	~\$0.002
Hallucination Rate	N/A	15% (Wrong tools)	5%	0% (Validated)
Correct Conflict / Pro Bono	70% / 50%	95% / 93%	88% / 85%	98% / 96%
Test 4 (Ambiguity) Result	Routed wrong area	Spun into 5 loops	Couldn't handle	Escalated correctly

Where It Stands Now

The **Constrained ReAct Agent** is our final, production-ready choice for Ashford & Kane LLP.

Next steps for production deployment:

- Increase our lab budget to MAX_STEPS=8 just for safety margin.
- Implement a "Human-in-the-Loop" override, allowing attorneys to intercept and reject automated routing.
- Ensure every automated conflict check is logged directly to our internal ethics database.
- Configure monitoring systems to trigger alerts if the system's escalation rate spikes above 20%.



Image Sources



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